Response to the editor and reviewers

Alexander V. Mantzaris

Editor letter

Editor Comments

"The manuscript presents a technically interesting and systematically evaluated framework for embedding cryptographic payloads into language-model-generated text through deterministic token-rank manipulation. Reviewer 2 considers the previous concerns largely addressed. However, several substantive issues remain before the conclusions can be fully supported. In particular, the authors should clarify the robustness boundaries under text transformations and configuration desynchronization, analyze the implications of zero cross-model recovery, explain the near-perfect detectability under an exact-model-aware attacker, strengthen semantic-coherence validation, document all deterministic filtering rules, and provide deeper analyses of entropy-gating failures, computational complexity, and quantization-induced divergence. Suggested references are optional, and the authors may select appropriate recent peer-reviewed literature." -Chun-Wei Yang

We recommend submitting all revisions within the mentioned deadline.

If you need more time, please contact us and include your submission ID.

Kind regards,

Purva Dhamankar —

We have revised the claims to distinguish concealment of surface form, exact configured recovery, visible-text transport, detectability and naturalness as perceived by readers. Adverse and inconclusive outcomes remain visible.

The V4 text corrects the historical raw and limited unmodified references to saved-token recovery, documents the entropy experiment with no filter and sampled skips, and separates the directional fixed-message quantization endpoint from free generation. The new contextual follow-up uses matched greedy controls and reports contextual coherence and acceptability under the common automated rubric. All earlier fragment outcomes remain in the supplement.

Main Results under Contextual acceptability of complete messages, Discussion, and Supplementary Notes S16 through S20. Accompanying documents, main4.pdf pages 11, 16, 19 and supplementary4.pdf pages 41, 42.

Code deposit.

Answer. The code and retained data for this V4 revision are available at https://github.com/mantzaris/llm-rankcloak. Published version 2.0.0, DOI 10.5281/zenodo.22555497, covers V3 only.

Change and evidence. The availability statements distinguish repository access to V4 from the published V3 archive. A new version under concept DOI 10.5281/zenodo.21987449 covering the current V4 code and evidence remains outstanding.

Location. Main Data Availability and Code Availability. Accompanying documents, main4.pdf pages 20.

Answer. The revision retains the theoretical and implementation analyses and limits the contribution to artifact-specific concealment of surface form and configured inversion. The completed study of intact messages supplies substantive automated evidence on contextual coherence and acceptability.

Change and evidence. The new study applies the same lenient, fixed rubric to complete encoded messages and matched greedy ordinary controls. It uses authentic prompts, blinded judges from other model families, a separate constructed calibration set and a frozen 48-payload sample. Among 537 complete message pairs in 280 trials, the equally weighted panel accepts 90.3 [88.4, 92.1]% of encoded messages and 96.6 [95.0, 98.1]% of matched ordinary messages. The paired difference is -6.3 [-8.7, -3.9] percentage points. Brackets give 95% payload-cluster intervals stratified by class. Mean logical connectedness is 3.60/5 for encoded text and 4.00/5 for ordinary text. Binary judge disagreement is 14.6% and 6.7%, respectively. The previous fragment assays are retained as supplementary methodological history, with their failed semantic control checks and limited local likelihood findings. Historical recovery, transport, entropy, quantization and detector results are unchanged. These automated findings do not establish confidentiality or indistinguishability.

Location. Abstract, contextual Methods and Results, Discussion and limitations, Supplementary Notes S17 and S20, and the answer to Reviewer 1 comment 4. Accompanying documents, main4.pdf pages 10, 16, 19, 20 and supplementary4.pdf pages 36, 42.

Reviewer 1 general assessment and wording instruction

> *Reviewer 1*
>
> *The manuscript proposes a steganographic method embedding cryptographic payloads into language model outputs via deterministic token rank manipulation. While demonstrating exact recovery under strictly controlled conditions, the framework exhibits fragility against text transformations and cross-model decoding, limiting its practical utility despite comprehensive evaluations of capacity and neural steganalysis detection.*
>
> *Overall, the study presents a reproducible measurement framework for linguistic steganography. However, its reliance on perfectly synchronized configurations and failure during standard markdown operations require deeper theoretical justification. Furthermore, near-perfect detectability by exact-model-aware attackers and the absence of perceptual validation weaken its broader claims. Significant revisions addressing robustness boundaries, theoretical limitations of detectability, and expanded discussions on quantization sensitivity are required before publication can be considered.*
>
> *Please do not arbitrarily simplify or rephrase the review comments, as this hinders the assessment of your revisions.*

Answer. We agree that recovery requires a synchronized configuration and that the observed transport and detection failures substantially restrict some of the practical claims. The revision

presents these as limitations honestly, not as all possible problems solved by this new robustness method.

Change The configuration sensitivity analysis states that a fully configured attacker may decode, distinguishes trace informed detection from visible-text scoring, and labels the block/ECC fallback as unimplemented. A new contextual evaluation of intact messages supplies automated evidence under an explicit lenient rubric. The unsuccessful earlier semantic pilots remain in Supplementary Note S17. The additional transport diagnostics are illustrative rather than population estimates.

Location. Main Results under Configuration sensitivity, detection and a proposed fallback, limitations, and Supplementary Notes S17 through S20. Accompanying documents, main.pdf pages 19, 20 and supplementary.pdf pages 36, 40, 41, 42.

Preservation of the review wording.

Answer. We have retained the complete review wording without simplification or paraphrase, including both general paragraphs and this instruction.

Change All thirteen original quoted blocks are retained, including the editor letter, Reviewer 2 acknowledgment, both Reviewer 1 introductory paragraphs and the instruction to preserve wording. Answers remain separate from the original comments.

Reviewer 1 comment 1

> *1 In the abstract and introduction, the manuscript admits severe transmission fragility. While you explicitly scope this as a measurement framework, you must expand the theoretical discussion on why standard operations like markdown copy cause complete failure and discuss potential boundary mitigations.*

Answer. The tested Markdown operation is a specific composite transform. It adds a blockquote prefix to each line and removes whitespace. The receiver then applies saved token offsets after retokenization. This can shift the forced span, change tokens and alter contextual ranks. It does not demonstrate failure for every Markdown workflow.

Change and evidence. Five sources selected by identity hash give 25 decomposed arms and 135 segment instances. Exact reuse leaves 26 missing segment replays. We also checked all 53 reused segment inputs at the new context allocation, with identical rank and error outcomes. These checks add no logical cover arms. Prefix-only insertion recovers none of the five. All 27 selected segment instances are rejected at a prohibited first forced token before a rank sequence is returned. These endpoints do not measure downstream rank amplification. Declared wrapper removal recovers one originally failing Mistral case, but cannot restore removed whitespace. Byte edits, extracted spans, first rank differences, errors and recovered bytes are retained. Original composite recovery remains 0/144.

Location. Main Methods under Decoder compatibility and bounded transport diagnostics, Results under Mechanisms of transmission failure, and Supplementary Note S18 with its illustrative recovery table. Accompanying documents, main4.pdf pages 10, 18 and supplementary4.pdf pages 40, 41.

Reviewer 1 comment 2

> *2- Within the bounded payload methods section, assuming perfectly synchronized model configurations is highly restrictive. Please introduce a theoretical discussion addressing how minor configuration desynchronization, such as slight tokenizer version updates, theoretically impacts the overall payload recovery rate.*

Answer. A minor version change has no universal implication for recovery rate. It may preserve the relevant tokenization or change token identities, offsets and conditioning. Model or gate differences introduce other disagreement mechanisms. We state the required shared contract and conditional agreement argument explicitly.

Change A new canonical configuration fingerprint includes eleven required contract components. Tests accept matching configurations, verify deterministic serialization and reject changes to every component. It is a compatibility API, not authentication or a robustness guarantee, and was not retroactively applied to the historical messages.

Location. Main Methods under Decoder compatibility and bounded transport diagnostics, main Results under Configuration sensitivity, detection and a proposed fallback, and Supplementary Note S19. Accompanying documents, main.pdf pages 10, 19 and supplementary.pdf pages 41.

Reviewer 1 comment 3

> *3 In the exact copy fragility evaluation, the zero percent recovery rate for cross-model decoding is a severe limitation. You should propose and discuss a conceptual error correction strategy or fallback mechanism to mitigate this lack of robustness during mismatched decoding.*

Answer. Zero cross model recovery remains a severe limitation. We propose configuration identification before decoding, independently framed blocks with context resets, integrity checks, bounded outer error/erasure correction and retransmission. This is a conceptual fallback, not an implemented improvement.

Change and evidence. The argument states the minimum-distance condition $2e+s<d$ and its assumptions. ECC cannot reconstruct an arbitrary wrong model codebook or repair persistent synchronization loss. Reliable framing, a supported configuration and overhead for redundancy and retries remain necessary. Retransmission under an unresolved mismatch is ineffective.

Location. Main Results under Configuration sensitivity, detection and a proposed fallback, and Supplementary Note S19. Accompanying documents, main4.pdf pages 19 and supplementary4.pdf pages 41.

Reviewer 1 comment 4

> *4 Throughout the segmented multicover method section, the distinction between payload-bearing spans and natural tails lacks perceptual validation. You must provide stronger automated semantic coherence metrics to prove that these generated transitions appear natural and logically connected to readers.*

Answer We have made contextual acceptability of intact messages the primary automated coherence evaluation. Two blinded local model judges assess each complete message in its authentic prompt

context without seeing the payload boundary. The fixed rubric accepts clear meaning and minor awkwardness at disruption levels 0 or 1, while levels 2 and 3 indicate material interference with understanding. The study was designed after the fragment results and frozen before its own scoring.

Change A seeded selection covers 48 payloads across four hexadecimal classes, three generators and two schedules, giving 576 messages outside the earlier scored cohort. Each is paired with ordinary output from the same generator and prompt using matched filtering, greedy continuation, budget and stopping rules. Among 537 complete pairs, panel acceptance is 90.3% for RankCloak and 96.6% for ordinary output. The paired difference is -6.3 percentage points with a 95% payload-cluster interval of [-8.7, -3.9]. Mean logical connectedness is 3.60/5 versus 4.00/5.

Messages are averaged within trials and payloads, with intervals conditional on the fixed reused controls. Supplementary Note S20 reports calibration mistakes, the 39 incomplete pairs, missing score sensitivity, subgroup results, disagreement and exact unfavorable examples. Judges and generators are not fully crossed, and the comparison does not isolate a causal boundary effect. This completed study provides automated evidence of contextual coherence and acceptability under the stated rubric. The earlier fragment pilots and limited likelihood evidence remain in S17. The automated findings address the requested stronger coherence evaluation.

Location Main Methods, Intact-message contextual acceptability, and Results, Contextual acceptability of complete messages, with Table 3 and Figure 5. Supplementary Note S20, Selection and matched generation, Complete judging rubric, Judge settings and calibration, Estimation and complete results, Stops, lengths, reuse and missingness, and Full-message examples. Note S17 retains the historical assays. Accompanying documents, main4.pdf pages 10, 16, 17 and supplementary4.pdf pages 36, 42, 44, 45, 46.

Reviewer 1 comment 5

> *5 Neural steganalysis results indicate near-perfect detection by the exact-model-aware attacker. Rather than a full redesign, you must deeply analyze the theoretical bottlenecks causing this glaring statistical detectability and clearly define the framework's security boundaries against this specific threat model.*

Answer The evaluated model-aware attack uses retained traces and exposes a strong statistical limitation. Its features come from saved token likelihoods under the generating model. Reproducing them requires the original token path, model/backend, prompts, boundaries, resets and probability convention. A scorer with access only to visible text is a distinct condition.

Change The added discussion explains that payload-rank distributions need not match ordinary sampling. It distinguishes untempered scores over the full vocabulary from the temperature-0.8 top-p-0.95 sampler and states the support conditions for a sequence-KL argument. AUC is not converted to KL, and no unique causal feature attribution is claimed. An attacker with the complete receiver configuration may decode. No confidentiality or indistinguishability against an informed attacker is asserted.

Location Main Neural steganalysis, statistical analysis, and computational overhead, Configuration sensitivity, detection and a proposed fallback, and Supplementary Notes S12 and S19. Accompanying documents, main4.pdf pages 9, 19 and supplementary4.pdf pages 23, 41.

Reviewer 1 comment 6

> *6 The discussion of entropy gating notes that the strict gate condition resulted in maximum budget payload completion failures. You should investigate and report the exact linguistic contexts or token-level characteristics that triggered these specific failures during the generation loop.*

Answer All six strict gate failures are retained and analyzed at their original budgets. Five are Qwen and one Mistral, all ASCII B16. They consume 110/128, 116/128, 54/64, 106/128, 59/128 and 116/128 ranks. Strict completion remains 114/120.

Change The supplement reports exact windows aligned with the pinned tokenizer, longest below-threshold runs of 50, 163, 39, 35, 267 and 51 positions, rolling progress, margins and rank pressure. The longest runs contain 30 to 157 distinct token IDs, at most three identical IDs consecutively, and rank-one frequencies of 85.7 to 98.8 percent. All 605 positions remain ordinary sampled skips with no payload progress. Twelve matched ungated/moderate trajectories and three strict successes selected by identity hash are retained. Observed contexts include formatted public service lists and project updates, but runs with low entropy also occur in successes. These are descriptive associations, not established linguistic causes. The trajectories used no token filter and sampled ordinary skips, not greedy skips.

Location Main Results under Strict-gate failure contexts and Supplementary Notes S13 and S16, including the six-case table, trajectory figure and exact local windows. Accompanying documents, main4.pdf pages 18 and supplementary4.pdf pages 26, 33, 35.

Reviewer 1 comment 7

> *7 The computational overhead analysis reports significant performance bottlenecks. Instead of exhaustive hardware baselines, please provide a theoretical computational complexity analysis comparing your rank transcoding method against standard linguistic steganography tools to mathematically justify the observed processing costs.*

Answer The new comparison separates model work from sorting, rank counting, probability passes and coding operations. Rank selection sorts all allowed candidates, whereas recovery of a known token counts greater logits and lower-ID ties.

Change The analysis inspects the primary arithmetic-coding and SAAC implementations and ADG grouping algorithm. It includes repeated diagnostic sorts, finite-precision cumulative tables, grouping search/removal, prompt prefills, KV state, logits_all storage, tails, sampled skips, segmentation and model passes over direct subwords on the payload side. Bounds are implementation-specific and imply no superiority in wall time across backends.

Location Main Computational overhead and generalizability and Supplementary Note S19, implementation specific operation count table. Accompanying documents, main4.pdf pages 18 and supplementary4.pdf pages 41, 42.

Reviewer 1 comment 8

> *8 In the reference list, citation 16 is an unverified preprint. If formally published, update the citation; otherwise, provide detailed theoretical justification for relying on its entropy-guided watermarking claims, or replace it with a peer-reviewed publication to strengthen your foundation.*

Answer The bibliography key Cai2025EntropyGuidedWatermarking refers to the previous reference 16, arXiv:2504.12108. The inspected primary record lists no journal reference. We have replaced its role in the V4 argument with the peer-reviewed Lee and colleagues ACL 2024 paper on entropy-threshold selective watermarking.

Change The V4 prose now cites Lee2024SWEET for the limited thresholding precedent. It distinguishes detecting a watermark from recovering a chosen payload. Neither a watermark theorem nor the unverified preprint is used as a RankCloak recovery guarantee. The historical experiment itself is still documented by its original configuration and data.

Location Main Introduction, Supplementary Note S13 and the newest version, V4 bibliography. Accompanying documents, main4.pdf pages 1 and supplementary4.pdf pages 26.

Reviewer 1 comment 9

> *9 The methods section describing the deterministic token filter leaves the rules for excluding markup fragments obscure. You must explicitly list these complete exclusion criteria in the main text so independent researchers can accurately replicate your experimental setup.*

Answer The main Methods now give the complete deterministic token filter specification, including every literal blocked substring and all other exclusion criteria. The historical filter is unchanged.

Change The specification covers empty and replacement pieces, ASCII controls and exceptions, whitespace/start rules, backslashes, individual-token decoding and exceptions, mask application, special-token distinctions and stable tie-breaking. It distinguishes full selection sorting from counted rank recovery and notes limitations for concatenated markup and nonfinite logits. The versioned source link and source hash are retained in the audit, with focused edge-case tests.

Location Main Methods under Complete deterministic token filter specification and its complete rule table. Accompanying documents, main4.pdf pages 5.

Reviewer 1 comment 10

> *10 The paired quantization experiment shows independently generated token paths diverging significantly despite identical inputs. Please expand this section to theoretically discuss how minor quantization errors in logits are amplified by the deterministic sorting process, triggering this avalanche effect.*

Answer We separate logit perturbations on the same history from subsequent feedback when independently generated token paths split. For a fixed history and allowed set, pairwise gaps above twice a uniform absolute perturbation bound preserve order. Smaller gaps and ties have no such protection. This is a conditional bound, not a measured quantization error distribution.

Change The audit reconciles 69,528 rank changes across 244,440 positions in 1,920 pairs. Encoded messages contribute 56,321/122,220 and ordinary controls 13,207/122,220. The distinct offline endpoint for fixed messages gives zero exact Q4-to-Q8 payload recoveries in 960 supported rows, with 766 trials with invalid ranks and 194 valid but wrong decoded payloads. It supports no claim in the reverse direction. Observed-token replay feeds received cover tokens, not decoded payload symbols, back into the model. Existing evidence supports these claims without new replay of neighboring logits.

Location Main Results under Fixed-message cross-quantization decoding and Supplementary Notes S14, S18 under Directional cross-quantization endpoint, and S19. Accompanying documents, main4.pdf pages 18 and supplementary4.pdf pages 29, 41.